Batch Lab: Scale a Recipe, Prove the Measure

Tracking #: 2026-11-13-NAT-COOK-WATER-BATCH-46

Water-only simulation; per-serving quantity is three quarters cup

Servings	Correct total water
2	One and a half cups
4	Three cups
6	Four and a half cups
8	Six cups
Lab note	Use only clean water and safe cups

1. Show a labelled scaling table for 2, 4, 6, and 8 servings. Write the equal-groups or multiplication reasoning for EACH total, not only final numbers.

2. Design TWO different level-cup pour plans to make four and a half cups. One must use full cups and a half-cup; the other must use quarter-cup or half-cup combinations. Check the totals.

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3. With safe clean tools, PHYSICALLY MEASURE a four-and-a-half-cup test batch using water. If the bowl is small, split it into two measured containers; record actual POUR counts and any changes.

4. Error audit: A student made four cups for four servings. Write what the per-serving amount was supposed to be, calculate the correct total, quantify the error and show the corrected measuring plan.

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5. Suppose one quarter-cup pour is accidentally doubled. Determine how much extra water that adds and write a safe, accurate correction plan. State whether this happened in your real trial or only in the scenario.

6. Complete real cleanup and a six-sentence lab report: recipe target, calculated amount, measuring method, actual observation, how you checked accuracy, and one improvement. Record actual setup, measuring and cleanup minutes separately.